Purrnanssh Sinha

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SKILLS

Languages: C++, Python, C, SQL, HTML, CSS
Frameworks/Tools: NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, Streamlit, MySQL, Excel, Power BI, Microsoft Azure
LLM & AI Systems: Retrieval-Augmented Generation (RAG), LangChain, Groq API, Whisper, HuggingFace Transformers, Sentence Transformers, Prompt Engineering
Core Concepts: ML Algorithms, NLP, Statistics, Embedding Models, Vector Databases
Soft Skills: Problem-Solving, Business Impact Assessment, Project Management, Communication

INTERNSHIPS

National Highways Authority of India | Data Science Intern Jun' 25 – Aug' 25

- Created a scalable ingestion pipeline with Delta Lake, PySpark, and Structured Streaming, enabling incremental, real-time data appends with robust version control and time zone-aware logic.
- Programmed synthetic data generators and automated ingestion scripts using Delta Table, maintaining end-to-end data consistency across versions.
- Executed an integrated email notification system in python with HTML-based insights, improving observability and reducing manual oversight.

PROJECTS

CodeMap | Python, TypeScript, CSS | GitHub Mar' 26

- Built an interactive codebase analysis platform that converts GitHub repositories into dependency graphs, enabling developers to visualize software architecture, module relationships, and code hotspots.
- Developed a static analysis engine in Python using AST parsing to extract function-level dependencies, fan-in/fan-out metrics, and complexity insights from large codebases.
- Engineered a high-performance React + TypeScript visualization layer using D3/Force Graph to render and explore complex software networks with interactive dependency inspection.

National Highways Intelligence Project | Python, LangChain, Whisper | GitHub May' 26

- Designed a Smart Toll Intelligence System integrating Weigh-In-Motion (WIM) sensors, ANPR-based vehicle recognition, and FASTag workflows to automatically detect overloaded commercial vehicles and calculate overload penalties.
- Architected with a scalable highway monitoring platform for real-time vehicle classification, axle detection, weight verification, and automated toll enforcement using Computer Vision and Intelligent Transportation Systems (ITS) concepts.
- Developed end-to-end system design including data pipelines, analytics dashboards, and revenue-recovery mechanisms aimed at reducing toll leakage, improving road safety, and supporting nationwide deployment across highway corridors.

CERTIFICATES/CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 AI Foundations Associate (1Z0-1122-24) | Oracle Feb' 25
- Oracle Cloud Infrastructure 2024 Generative AI Professional (1Z0-1127-24) | Oracle Jun' 24
- Privacy and Security in Online Social Media | NPTEL | Link May' 24

ACHIEVEMENTS

- **Achieved 3 Star SQL proficiency on HackerRank** through hands-on problem solving in relational databases and query optimization.
- **Solved 290+ Data Structures & Algorithms** problems across platforms, including 100+ on LeetCode.

EDUCATION

Lovely Professional University Phagwara, Punjab
Bachelor of Technology Aug' 23 – Present
Computer Science and Engineering